

AI Code Review Report

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Overall Quality Score

54/100

Score Breakdown

Metric	Score
Readability	40/100
Security	50/100
Performance	84/100
Maintainability	50/100
Documentation	50/100

Complexity: Medium

Issues Found

Bugs

- Line 164: Potential division by zero (*critical*)

Fix: Check if denominator is zero before division

Security Issues

- Line 16: Hardcoded credentials found - use environment variables (*high*)

Remediation: Use environment variables or secret management

- Line 46: Potential SQL Injection vulnerability - use parameterized queries (*critical*)

Remediation: Use parameterized queries (e.g., `cursor.execute("SELECT * FROM users WHERE id=?", (user_id,))`)

- Line 52: Potential Command Injection - avoid user input in system calls (*critical*)

Remediation: Use subprocess with list arguments instead of `shell=True`

- Line 84: Insecure deserialization - avoid pickle for untrusted data (*high*)

Remediation: Use JSON or other safe serialization formats

- Line 89: Weak cryptographic algorithm - use stronger alternatives (*high*)

Remediation: Use `bcrypt`, `argon2`, or `PBKDF2` instead of MD5/SHA1

Performance Issues

- Line 96: Deeply nested loops (4 levels) (*medium*)
- Line 112: Using `range(len())` - consider using `enumerate()` (*low*)

Style Issues

- Line 31: Missing docstring
- Line 32: Missing docstring
- Line 43: Missing docstring
- Line 51: Missing docstring
- Line 55: Missing docstring
- Line 59: Missing docstring
- Line 66: Magic number 50 found
- Line 70: Missing docstring
- Line 71: Magic number 42 found
- Line 78: Missing docstring
- Line 82: Missing docstring
- Line 88: Missing docstring
- Line 93: Missing docstring
- Line 103: Missing docstring
- Line 106: Magic number 1000000 found

- Line 110: Missing docstring
- Line 118: Line is 129 characters (max 100)
- Line 118: Missing docstring
- Line 120: Magic number 1000 found
- Line 122: Magic number 50 found

... and 66 more style issues

Suggestions & Fixes

Code Refactoring Recommendations

Based on the analysis

- • ■ Fix 4 bugs
- • ■ Optimize 5 performance issues
- • ■ Improve 56 style issues

General Suggestions

- Fix Critical Bugs: Found 1 bugs
- Security Vulnerabilities: Found 8 security issues
- Performance Optimizations: Found 5 performance issues
- Code Style Improvements: Found 86 style issues

Specific Fixes

Bug Fix

Original: Line 164

Fixed: Check if denominator is zero before division

Explanation: Potential division by zero

Security Fix

Original: Line 16

Fixed: Use environment variables or secret management

Explanation: Hardcoded credentials found - use environment variables

Security Fix

Original: Line 17

Fixed: Use environment variables or secret management

Explanation: Hardcoded credentials found - use environment variables